DIGITAL FORENSICS CASE REPORT

Document Provenance and Movement

Independent forensic examination of the Justine Beaufort training case in Magnet AXIOM Examine

Prepared by Hadis Pazhand | Training dataset (fictional) | All times are UTC+0:00 unless stated

ABOUT THIS REPORT

This report documents a self-directed examination of a fictional academic training case (Justine Beaufort), supplied for a university digital forensics unit. All names, accounts and events are fictional. The findings describe what the artifacts show. They do not establish who was using the computer or why. Source screenshots are cited by figure ID (for example, P1-03); the main figures are in the body and the supporting figures are in Appendix A.

1 Summary

The artifacts support the following account of the afternoon of 26 March 2019:

- **Two documents were downloaded through Chrome shortly after file-type searches.** `Whooo_Am_I_Owls_Bro.pdf` was downloaded at 3:31:33 PM, 11 seconds after a search for "owls filetype:pdf" and 7 seconds after a Chrome visit to the PDF's address. `Thoughts-on-Ambassador-Owls.docx` was downloaded at 3:49:18 PM, 4 to 5 seconds after a search for "owls filetype:docx".
- **A copy of the PDF was created on a removable drive** (F:, volume serial number FC9BD2D2) at 3:31:37 PM, inside the download window.
- **The same PDF was emailed to Jim Turk at 3:48:40 PM.** The attachment's MD5 and SHA1 hashes match the PDF on disk.
- **The removable drive was in use from about 2:49 PM to 4:49 PM.** Shortcut and Jump List records reference at least six files on it, attributed by AXIOM to Edge, Photos and Excel. The device's own registry record shows only a 37-second session at 12:40 PM. This discrepancy is unresolved.
- **No cloud-storage record was located** (for example Dropbox), and no email attachment carrying the names of the Word document, `Great Horned Owl.pdf` or `owlColors_backup.xlsx` was found.
- **One timestamp is unexplained.** The Downloads copy of the PDF shows a file system Created time of 3:45:00 PM, about 13 minutes after the download and later than its Modified time (3:31:40 PM).

2 Objective and questions

Objective. To establish, from the artifacts in the Justine Beaufort training case, how two documents (`Whooo_Am_I_Owls_Bro.pdf` and `Thoughts-on-Ambassador-Owls.docx`) reached the computer, what searches and page visits preceded them, where copies went afterwards (removable media and email), and what other activity surrounded those events, and to present the findings as a verified timeline.

- **Q1 Provenance.** Where did each document come from, how, when and where was it saved?
- **Q2 Lead-up.** What was searched and viewed just before each download?
- **Q3 Removable media.** What is on device FC9BD2D2, when was it connected, and was anything opened from it?
- **Q4 Sharing.** Did any of the documents leave the machine by email, cloud storage, social media or chat?
- **Q5 Context.** What else was happening on the machine around each event?

Two further files on the drive (`Great Horned Owl.pdf` and `owlColors_backup.xlsx`) are examined under Q3 and Q4, but how they reached the computer is not established.

3 Scope and method

Case	“lecture” case, Magnet AXIOM Examine v8.9.1.43258
Evidence item	JustineBeaufort.E01, Partition 2 (Microsoft NTFS, 199.9 GB); 196,954 evidence items
Access	AXIOM Examine only. The case was already processed; no evidence was acquired or reprocessed.
Time zone	AXIOM displays UTC+0:00. Local time appears to be four hours behind (the email header shows −0400 and the USB record’s local-time field is four hours earlier).
Verification rule	Each key finding is supported by at least two independent artifacts where available. Where the artifacts do not prove a point, the wording is “consistent with”.

- Searched each filename and reviewed the results by category (Web Related, Operating System, Documents, Email & Calendar).
- Sorted Google Searches by Date/Time and reviewed the minutes before each download; searched the source domains for page-visit records.
- Searched the USB volume serial number (FC9BD2D2) and reviewed USB Devices, LNK, Jump List and Prefetch records.
- Compared the emailed attachment’s hash with the file on disk, and used the Timeline view to check the cluster of events at 3:31 PM.
- Duplicate rows sharing one timestamp, which are common in browser history, were counted once.

Guided and independent work. The course exercises supplied the starting points for the provenance of the two downloaded documents and the identity of the USB device. The search-to-download linkage, page-visit analysis, USB activity analysis, email and hash verification, and the unified timeline were carried out independently.

4 Findings

4.1 Q1: Where the documents came from

Whooo_Am_I_Owls_Bro.pdf

- **Download.** The Chrome Downloads record shows a source on dfw.state.or.us (`/wildlife/living_with/docs/`), start 3:31:33 PM, end 3:31:38 PM, saved to `C:\Users\Justine B\Downloads\Whooo_Am_I_Owls_Bro.pdf`, state Download Complete, opened by user: Yes (Figure P1-03).
- **Copy on removable media.** A shortcut (LNK) record shows the file on `F:\` (drive type DRIVE_REMOVABLE, volume serial number FC9BD2D2). The target file was created at 3:31:37 PM and last modified at 3:31:40 PM, and the shortcut was updated at 4:09:39 PM (Figure P1-05). The copy was therefore created one second before Chrome’s recorded end of the download, which is consistent with it being copied to the USB immediately after the download. Chrome’s end time can lag actual completion, and the mechanism of the copy is not established.
- **Document properties.** AXIOM shows Creator Adobe InDesign CC 2015 (Windows), Producer Adobe PDF Library 15.0, Created 5/16/2016 8:50:26 PM and Modified 5/17/2016 10:39:15 PM. These values are written by the authoring software and can be altered, and their time zone is not stated. They indicate the document was authored about three years before the download and do not show when it reached this computer (Figure P1-04). The preview shows an Oregon Department of Fish and Wildlife “Facts for Kids” owl sheet.

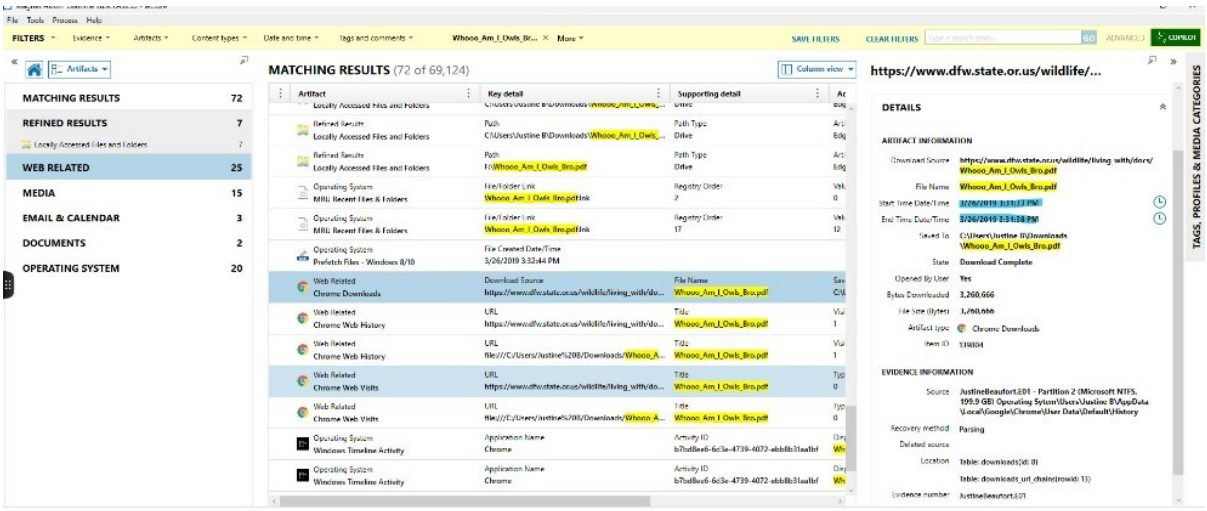


Figure P1-03. *Chrome Downloads record for Whooo_Am_I_Owls_Bro.pdf: source, start 3:31:33 PM, end 3:31:38 PM, saved to the Downloads folder, state Download Complete, opened by user Yes.*

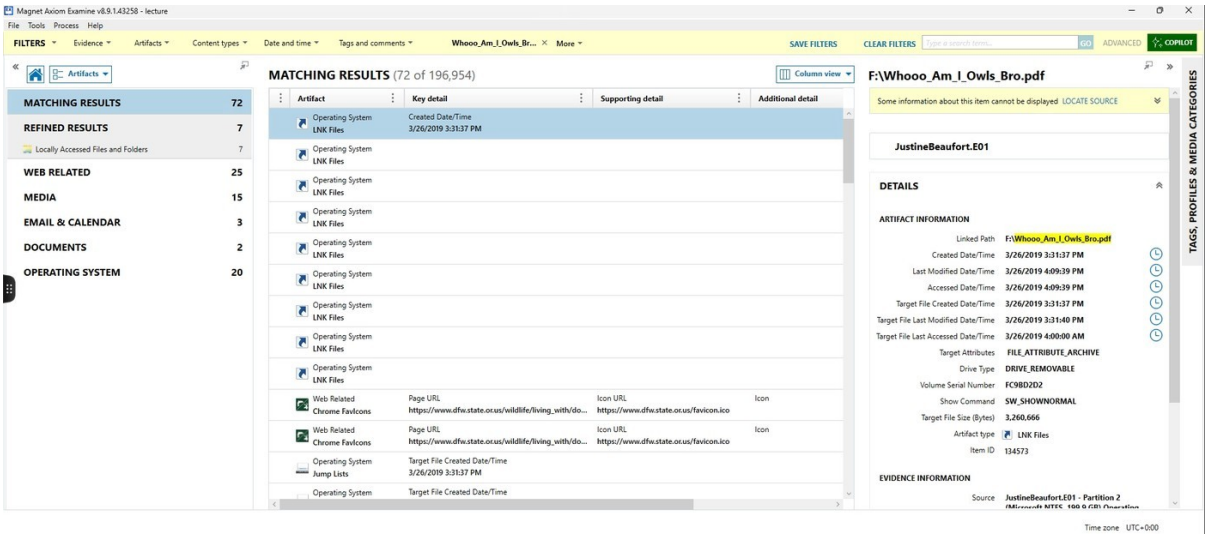


Figure P1-05. *LNK record for F:\Whooo_Am_I_Owls_Bro.pdf: drive type DRIVE_REMOVABLE, volume serial number FC9BD2D2, target file created 3:31:37 PM and last modified 3:31:40 PM.*

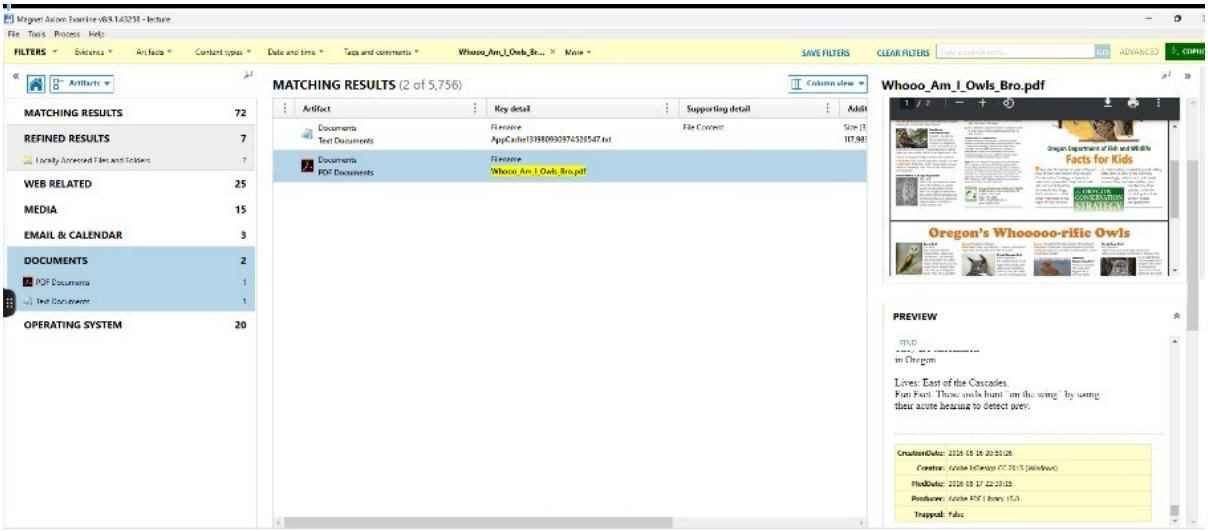


Figure P1-04. *PDF Documents record with preview and document properties (creator Adobe InDesign CC 2015, producer Adobe PDF Library 15.0, created 5/16/2016, modified 5/17/2016).*

Thoughts-on-Ambassador-Owls.docx

- **Download.** The Chrome Downloads record shows source `http://naturalencounters.com/site/wp-content/uploads/2018/01/Thoughts-on-Ambassador-Owls.docx`, start 3:49:18 PM, end 3:49:20 PM, saved to `C:\Users\Justine B\Downloads\Thoughts-on-Ambassador-Owls.docx`, state Download Complete, opened by user: Yes. Bytes downloaded and file size are both 151,474 (Figure P1-07). The record was parsed from the Chrome History database. A Chrome Cache Records entry for the same URL was first visited at 3:49:18 PM.
- **File system corroboration.** The file's Created (3:49:18 PM), Last Accessed (3:49:18 PM) and Last Modified (3:49:20 PM) times match the start and end of the Chrome record. These come from the NTFS file system, which makes them independent of the Chrome history (Figure P1-08).
- **Document properties.** Author and Last Author are both Steve Martin, with Created and Last Modified both 9/23/2017 8:34:00 AM. The preview title block reads "Some Thoughts on Ambassador Owls, Steve Martin, Natural Encounters, Inc.", which matches the source domain.
- **Limitation.** No LNK or Jump List item matched this filename, so "opened by user: Yes" rests on the Chrome record alone.

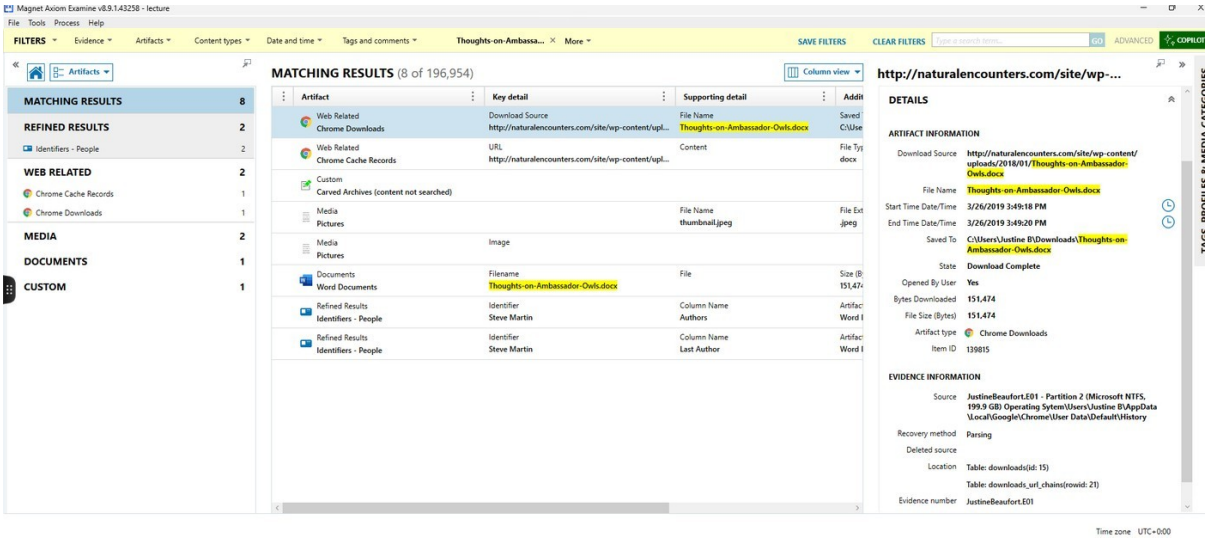


Figure P1-07. *Chrome Downloads record for Thoughts-on-Ambassador-Owls.docx (naturalencounters.com): start 3:49:18 PM, end 3:49:20 PM, 151,474 bytes, Download Complete, opened by user Yes.*

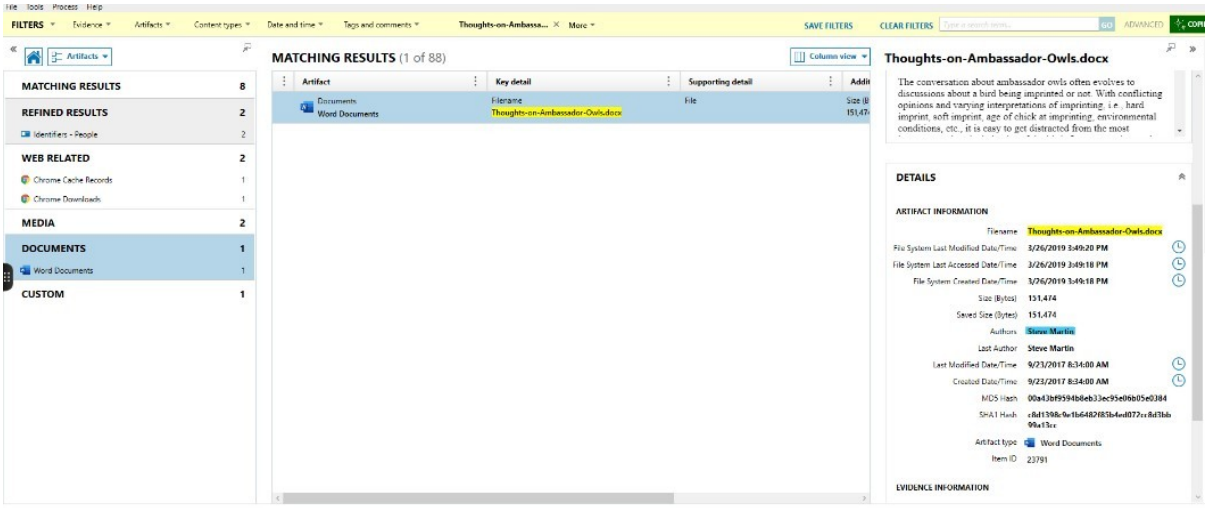


Figure P1-08. *Word Documents record: file system created 3:49:18 PM, accessed 3:49:18 PM, modified 3:49:20 PM; document properties show author Steve Martin and created and last modified 9/23/2017 8:34:00 AM.*

4.2 Q2: What came before each download

Time (PM)	Google search	Relation to downloads
3:29:39	owl documents	Before the PDF download
3:29:48	owl documents excel	Before the PDF download
3:29:58	wise owl tutorials	Before the PDF download; meaning not established
3:30:11	search by file type	Before the PDF download
3:30:21 to 3:30:51	search by file type google	Before the PDF download
3:31:22	**owls filetype:pdf**	**11 seconds before the PDF download began (3:31:33)**
3:31:47, 3:33:00	owls filetype:pdf (repeated)	After the PDF download
3:33:10	owls filetype:xlsx	After the PDF download; not matched to a download
3:46:36	owls	Before the Word download
3:46:41 to 3:47:24	owls funny	Before the Word download
3:49:13 to 3:49:14	**owls filetype:docx**	**4 to 5 seconds before the Word download began (3:49:18)**

Earlier searches for “pinterest” (3:17:34 PM) and “flickr” (3:21:51 PM), and Google Maps lookups for local places (3:26:57 to 3:29:01 PM), have no link to either download. The Details pane shows the search records were parsed from Chrome Web Visits, so one search can appear as several rows (Figure P2-03; supporting figures P2-01, P2-01b, P2-02 and P2-04 are in Appendix A).

Page visits. For the PDF, Chrome Web Visits records a visit to the PDF’s own address at 3:31:26 PM, with transition type LINK, typed count 0 and visit source Local. A cache entry for the same URL was first created at 3:31:25 PM. The sequence is therefore search (3:31:22), page visit (3:31:26), download (3:31:33) (Figure P2-05). For the Word document, only a cache entry (first visited 3:49:18 PM) and the download record were found, with no separate visit record, so that chain is partial (Figure P2-06 in Appendix A).

Finding. In both cases a `filetype:` search was run seconds before the download, and the earlier queries moved from “search by file type” to the `filetype:` operator. This is consistent with someone searching for documents by file type. The records show the queries and times only. They do not show intent or which search result was selected.

EVIDENCE (373)

Column view

Search Term	URL	Date/Time	Date...	Original Search Query
wise owl tutorials	https://www.google.com/search?rlz=1C1CHBF_enUS...	3/26/2019 3:29:58 PM		
wise owl tutorials	https://www.google.com/search?rlz=1C1CHBF_enUS...	3/26/2019 3:29:59 PM		
search by file type	https://www.google.com/search?rlz=1C1CHBF_enUS...	3/26/2019 3:30:11 PM		search by file type
search by file type	https://www.google.com/search?rlz=1C1CHBF_enUS...	3/26/2019 3:30:11 PM		search by file type
search by file type	https://www.google.com/search?rlz=1C1CHBF_enUS...	3/26/2019 3:30:12 PM		search by file type
search by file type google	https://www.google.com/search?rlz=1C1CHBF_enUS...	3/26/2019 3:30:21 PM		search by file type google
search by file type google	https://www.google.com/search?rlz=1C1CHBF_enUS...	3/26/2019 3:30:51 PM		search by file type google
search by file type google	https://www.google.com/search?rlz=1C1CHBF_enUS...	3/26/2019 3:30:51 PM		search by file type google
search by file type google	https://www.google.com/search?rlz=1C1CHBF_enUS...	3/26/2019 3:30:51 PM		search by file type google
owls filetype:pdf	https://www.google.com/search?rlz=1C2CHBF_enUS...	3/26/2019 3:31:22 PM		owls filetype:pdf
owls filetype:pdf	https://www.google.com/search?rlz=1C2CHBF_enUS...	3/26/2019 3:31:47 PM		owls filetype:pdf
owls filetype:pdf	https://www.google.com/search?rlz=1C2CHBF_enUS...	3/26/2019 3:33:00 PM		owls filetype:pdf
owls filetype:pdf	https://www.google.com/search?rlz=1C2CHBF_enUS...	3/26/2019 3:33:00 PM		owls filetype:pdf
owls filetype:pdf	https://www.google.com/search?rlz=1C2CHBF_enUS...	3/26/2019 3:33:00 PM		owls filetype:pdf
owls filetype:pdf	https://www.google.com/search?rlz=1C2CHBF_enUS...	3/26/2019 3:33:00 PM		owls filetype:pdf
owls filetype:xlsx	https://www.google.com/search?rlz=1C2CHBF_enUS...	3/26/2019 3:33:10 PM		owls filetype:xlsx
owls filetype:xlsx	https://www.google.com/search?rlz=1C2CHBF_enUS...	3/26/2019 3:33:10 PM		owls filetype:xlsx
owls filetype:xlsx	https://www.google.com/search?rlz=1C2CHBF_enUS...	3/26/2019 3:33:10 PM		owls filetype:xlsx
owls	https://www.google.com/search?q=owls&rlz=1C1C...	3/26/2019 3:46:36 PM		owls
owls	https://www.google.com/search?q=owls&rlz=1C1C...	3/26/2019 3:46:36 PM		owls
owls	https://www.google.com/search?q=owls&rlz=1C1C...	3/26/2019 3:46:36 PM		owls
owls	https://www.google.com/search?q=owls&rlz=1C1C...	3/26/2019 3:46:37 PM		owls
owls funny	https://www.google.com/search?rlz=1C1CHBF_enUS...	3/26/2019 3:46:41 PM		owls funny
owls funny	https://www.google.com/search?rlz=1C1CHBF_enUS...	3/26/2019 3:46:41 PM		owls funny

Figure P2-03. *Google Searches sorted by Date/Time, 3:29:58 to 3:46:41 PM, including "owls filetype:pdf" at 3:31:22 PM, 11 seconds before the PDF download began.*

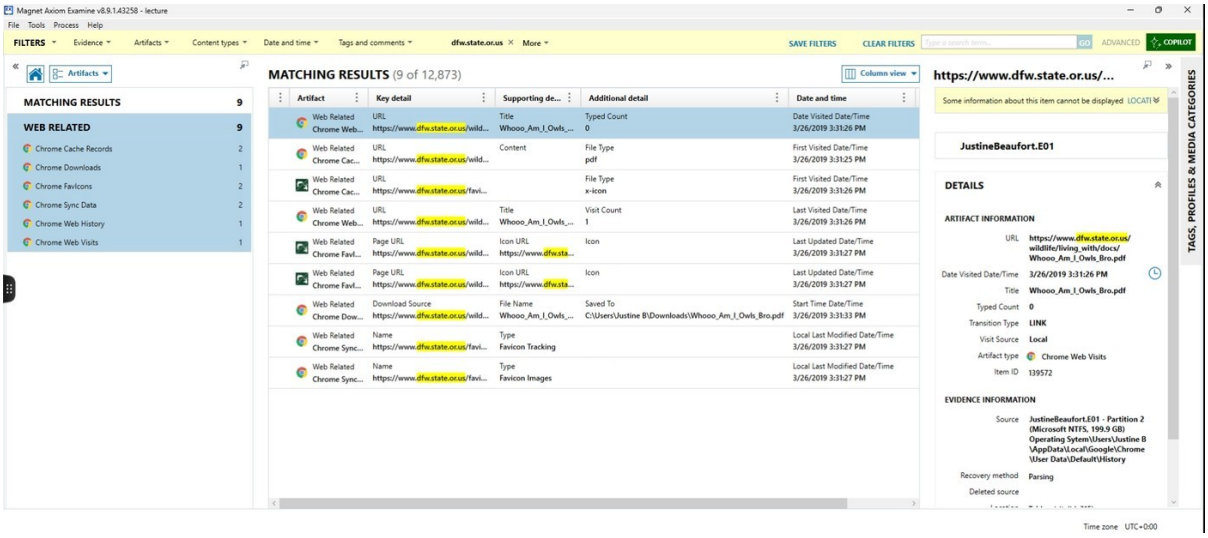


Figure P2-05. *Chrome records for dfw.state.or.us: a Web Visits entry for the PDF at 3:31:26 PM (transition type LINK, typed count 0) alongside the cache, favicon and download records.*

4.3 Q3: The removable drive

Friendly name	General UDisk USB Device
Device class ID	Disk&Ven_General&Prod_UDisk&Rev_5.00
Serial number	6&2017aed3&0&_&0
Associated user account	Justine B
Volume serial number	FC9BD2D2 (decimal 4238070482), matching the shortcut for the PDF copy on F:\
First install / last insertion	3/26/2019 12:40:21 PM
Last connected	3/26/2019 12:40:56 PM
Last removal	3/26/2019 12:40:58 PM
Source	SYSTEM registry hive

The volume serial number links the USB Devices record to the shortcuts for files on `F:\`. The association with the account Justine B ties the device to a user profile, not to a person (Figure P3-01).

Files on the drive. Searching the volume serial number returned 52 records, including 32 LNK files, 11 Jump Lists and 3 Prefetch files. They reference at least six files:

File on F:\	Target file created	Program (AXIOM's inference)
814u1IDExtL._SL1343_.jpg	2:49:06 PM	Photos
00y0y_j53rBCrjb8J_1200x900.jpg	2:50:24 PM	Photos
30588_112467978790101_5187737_n.jpg	2:54:30 PM	Photos
Whooo_Am_I_Owls_Bro.pdf	3:31:37 PM	Edge
Great Horned Owl.pdf	3:32:09 PM	Edge
owlColors_backup.xlsx	4:49:23 PM (last modified 4:48:16 PM)	Excel 2010

- **Reopened.** Shortcuts for `Great Horned Owl.pdf` and two of the images were updated between 3:42:32 and 3:42:46 PM, and the shortcut for the Whooo PDF at 4:09:39 PM. This is consistent with the files being opened again (Figure P3-03).
- **The spreadsheet.** Four artifacts place the opening of `owlColors_backup.xlsx` between 4:49:34 and 4:49:36 PM: Prefetch (Excel last run 4:49:34 PM), Windows Timeline Activity (Excel 2010 opening `F:\owlColors_backup.xlsx` at 4:49:36 PM), a RecentDocs entry in the Justine B profile (4:49:36 PM) and the shortcut. Its last-modified time (4:48:16 PM) is earlier than its created time on F:\ (4:49:23 PM), which is consistent with a copy. The source of the copy is not identified. Prefetch also lists Microsoft PDF Reader (last run 5:14:09 PM) and Photos (3/27/2019 12:06:11 PM) among the programs referencing the volume.
- **Paths.** Every path found for these files points to F:. No path outside F:\ appeared in the LNK, Windows Timeline Activity, RecentDocs or Edge/IE history records. Edge/IE history contains entries for files on F:/ (not itemised here).

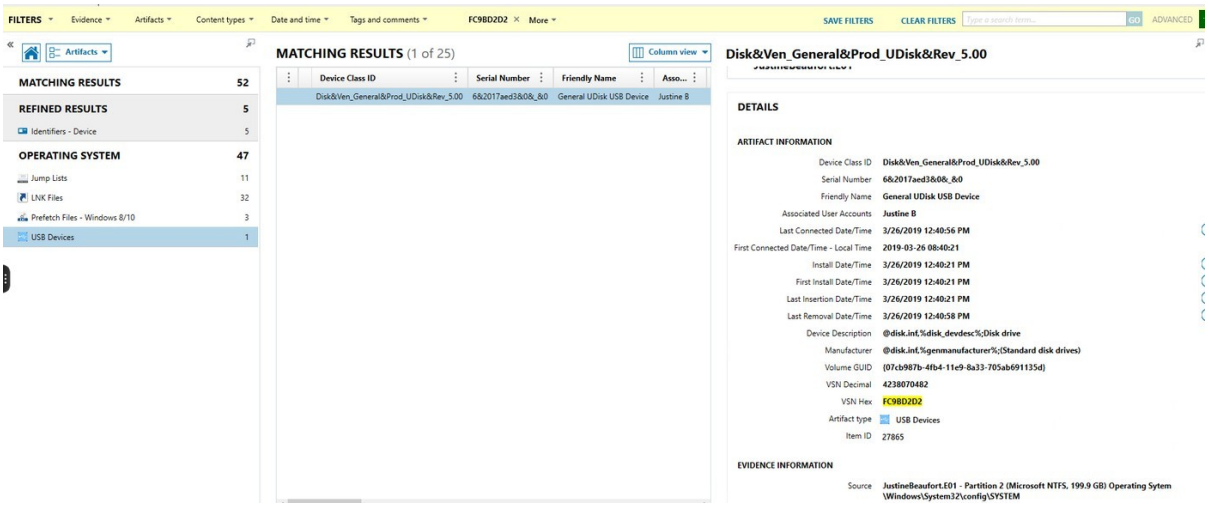


Figure P3-01. *USB Devices record: General UDisk USB Device, associated user Justine B, VSN hex FC9BD2D2, with one session recorded at 12:40 PM.*

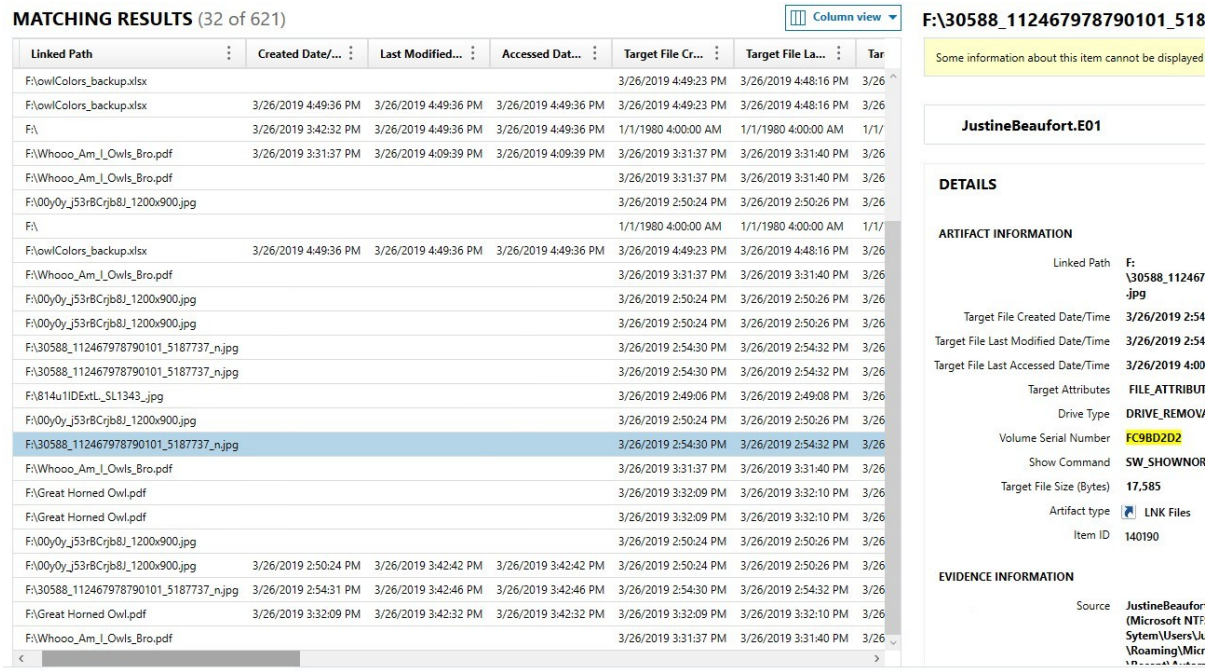


Figure P3-03. *LNK Files matching volume FC9BD2D2: files on F:\ with created, modified and accessed times from 2:49 PM to 4:49 PM.*

4.4 Q4: Whether anything was shared

Email. The PDF was attached to an email sent from the account owlgurl90@gmail.com (Justine Beaufort) to jimturk420@gmail.com (Jim Turk):

Subject and body	“Docs”: “Here are a couple of good docs.”
Submitted and delivered	3/26/2019 3:48:40 PM (header: Tue, 26 Mar 2019 11:48:40 −0400), about 17 minutes after the download ended
Attachments	Whooo_Am_I_Owls_Bro.pdf; BRF Bird Checklist, May2014 re c.members.xlsx
Location	owlgurl90@gmail.com.pst: Sent Mail and All Mail (two copies, one Message-ID)

Hash verification. The attachment’s MD5 (`d7cfcc3a9a3e68326ad877636f8a7bf0`) and SHA1 (`0d09f111c88f95ed580418d4cc10bbffeea07317`) match those of the PDF on the disk image, and its size (3,260,666 bytes) matches the size of the target recorded in the F:\ shortcut. The emailed file is therefore identical in content to the file on disk. Identical hashes do not show which copy was attached or who was at the

keyboard, and the records do not show that the recipient opened the message (Figures P4-02 and P4-08). The email records carry a Created time of 3/27/2019, about a day after sending. This is not explained, so the Submitted and Delivered times are used.

Document	Emailed?	Basis
Whooo_Am_I_Owls_Bro.pdf	Yes, 3:48:40 PM	Outlook PST (above)
Thoughts-on-Ambassador-Owls.docx	None found	The filename search returned no Email & Calendar results
Great Horned Owl.pdf	None found	“Great Horned” returned 13 email items. None of the 5 attachments carries that name. The items were not reviewed individually.
owlColors_backup.xlsx	None found	The filename search returned no Email & Calendar results
owls.pdf (separate file)	Yes, 2/19/2019 5:14:02 PM	Email “Owl Documents” to jimturk420@gmail.com with owls.pdf and WHM - Owls.pdf attached (Appendix A, P4-06 and P4-07)

- **Cloud storage.** A search for `owls.pdf` returned no Cloud category, so no Dropbox or other cloud-storage record was located in this case.
- **Separate activity.** The 2/19/2019 email is five weeks before the events above and is kept out of the 26 March timeline. A local copy at `C:\Users\Justine B\Documents\owls.pdf` appears in Edge/IE history at 5:14:08 to 5:14:09 PM on 3/26/2019 (carved from WebCacheV01.dat, so less reliable; Appendix A, P4-06b).

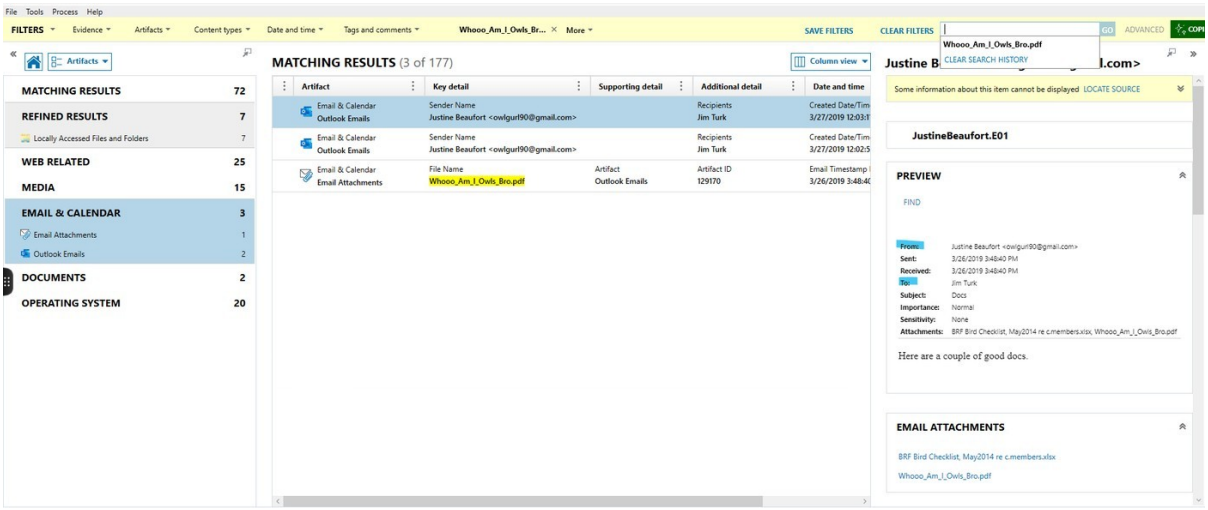


Figure P4-02. *Outlook email, Sent Mail copy: from owlgurl90@gmail.com to jimturk420@gmail.com, subject “Docs”, delivered 3/26/2019 3:48:40 PM, with Whooo_Am_I_Owls_Bro.pdf attached.*

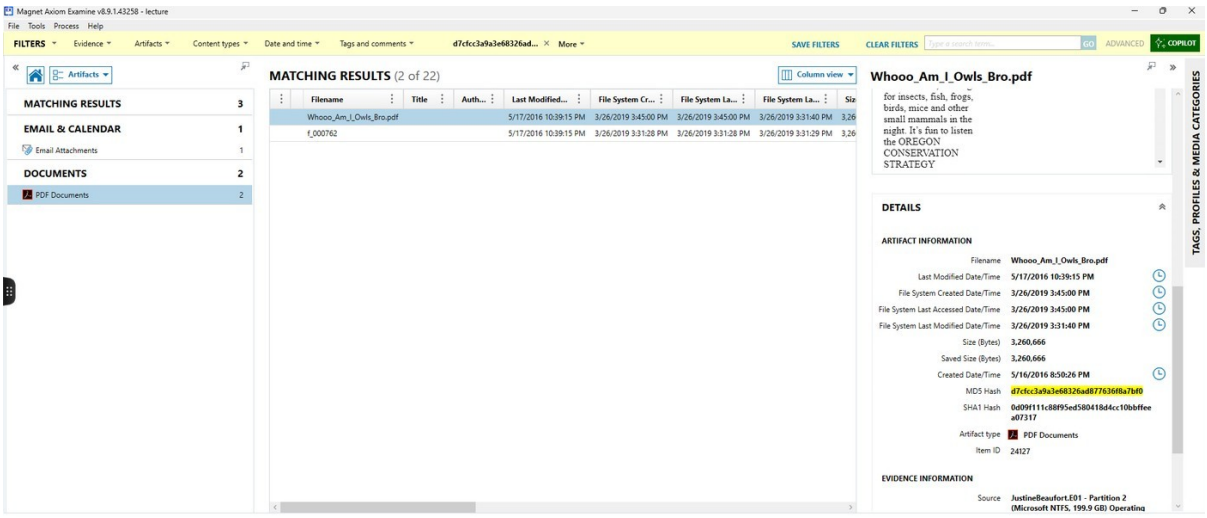


Figure P4-08. *Hash comparison: the MD5 of the emailed attachment (d7cfcc3a…7bf0) matches the PDF on disk; size 3,260,666 bytes.*

5 Unified timeline (Q5)

All times are 3/26/2019 PM, UTC+0:00, unless stated. Figure IDs point to the supporting screenshot.

Time	Event	Source	Figure
12:40:21 to 12:40:58	USB device FC9BD2D2: the only session in its registry record	USB Devices	P3-01
2:49:06	814u1IDExtL._SL1343_.jpg created on F:\	LNK	P3-03
2:50:24	00y0y_j53rBCrjb8J_1200x900.jpg created on F:\	LNK	P3-03
2:54:30	30588_112467978790101_5187737_n.jpg created on F:\	LNK	P3-03
3:29:39 to 3:30:51	Searches: owl documents; owl documents excel; search by file type	Google Searches	P2-02, P2-03
3:31:22	Search “owls filetype:pdf”	Google Searches	P2-03
3:31:25 to 3:31:26	Cache entry created and Chrome visit to the PDF’s address (link, not typed)	Chrome Cache, Web Visits	P2-05
3:31:28 to 3:31:29	Second PDF record f_000762 (same size, matched the hash search; likely a browser cache copy, path not examined)	PDF Documents	P4-08
3:31:33 to 3:31:38	**PDF downloaded to the Downloads folder**	Chrome Downloads	P1-03, P5-01
3:31:37	Copy of the PDF created on F:\	LNK	P1-05
3:32:09	Great Horned Owl.pdf created on F:\	LNK, Jump Lists	P3-03
3:33:00 to 3:33:10	Searches: owls filetype:pdf (repeated); owls filetype:xlsx	Google Searches	P2-03
3:42:32 to 3:42:46	Shortcuts for F:\ items updated (opened again)	LNK	P3-03
3:45:00	Downloads copy of the PDF shows file system Created and Accessed (later than its Modified time; cause not established)	PDF Documents	P4-09
3:46:36 to 3:47:24	Searches: owls; owls funny	Google Searches	P2-04
3:48:40	**PDF emailed to Jim Turk**	Outlook PST	P4-02

Time	Event	Source	Figure
3:49:13 to 3:49:14	Search “owls filetype:docx”	Google Searches	P2-04
3:49:18 to 3:49:20	**Word document downloaded** (cache entry 3:49:18)	Chrome Downloads, Cache	P1-07, P2-06
4:09:39	Shortcut for the F:\ PDF updated (reopened)	LNK	P1-05
4:45:40	Search “species of owls”	Google Searches	P2-04
4:49:23	owlColors_backup.xlsx created on F:\ (last modified 4:48:16)	LNK	P3-03
4:49:34 to 4:49:36	Excel runs and opens owlColors_backup.xlsx from F:\	Prefetch, Timeline Activity, LNK, RecentDocs	P3-05, P3-06, P3-07
5:14:08 to 5:14:09	C:\Users\Justine B\Documents\owls.pdf opened (carved record)	Edge/IE history	P4-06b

The related email of 2/19/2019 5:14:02 PM (`owls.pdf` and `WHM - Owls.pdf` to Jim Turk) is separate earlier activity and is not part of this timeline (Appendix A, P4-07).

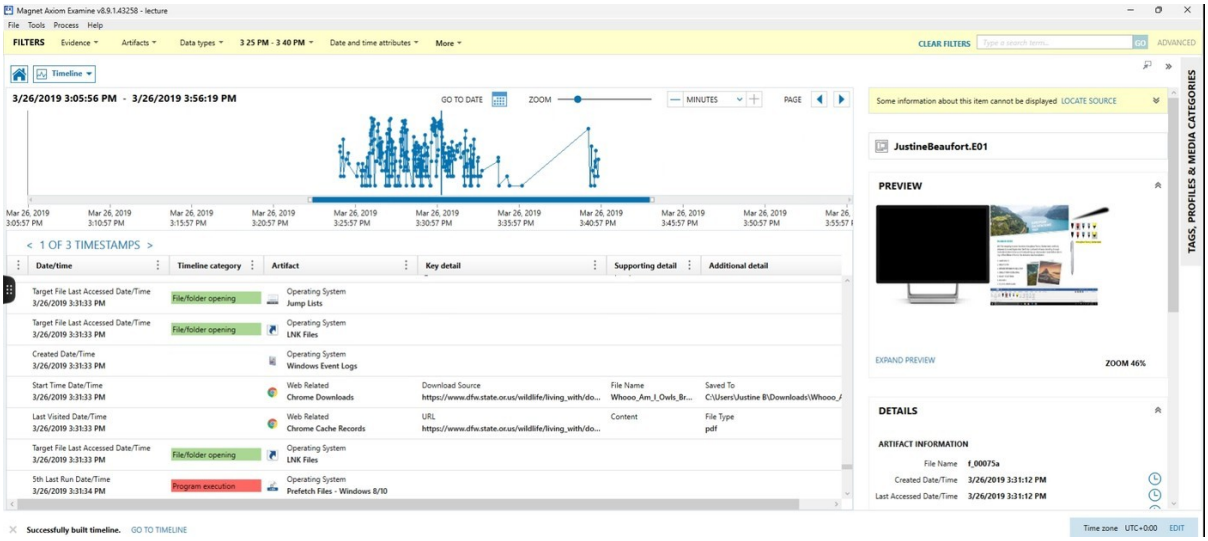


Figure P5-01. *Timeline view zoomed to minutes: records from several artifact types cluster at 3:31:33 to 3:31:34 PM, including the Chrome download start and the cache entry for the same URL.*

6 Discrepancies and open questions

6.1 Registry record against later use of the drive. The USB Devices record shows one session, 12:40:21 to 12:40:58 PM. Shortcut, Jump List and Prefetch records show files on the drive being created and opened from 2:49 PM to 4:49 PM. The device record therefore does not cover the later use. One possible explanation, not tested, is that the registry hive in the image does not reflect later changes. The cause is not established, and the conclusion that the drive was in use later rests on the shortcut, Jump List, Prefetch, Timeline Activity and RecentDocs records.

6.2 The 3:45:00 PM Created time. The Downloads copy of the PDF (Item ID 24127) has the same MD5 and SHA1 as the emailed attachment and the same size, but its file system Created and Last Accessed times are 3:45:00 PM, while Chrome records the download ending at 3:31:38 PM and the file’s Modified time is 3:31:40 PM. A Created time later than the Modified time is commonly seen when a file is copied, and both times fall exactly on the minute. Whether other files share the 3:45:00 PM time, which would point to a batch process, was not tested. The finding that the emailed file matches the file on disk is unaffected (Figure P4-09).

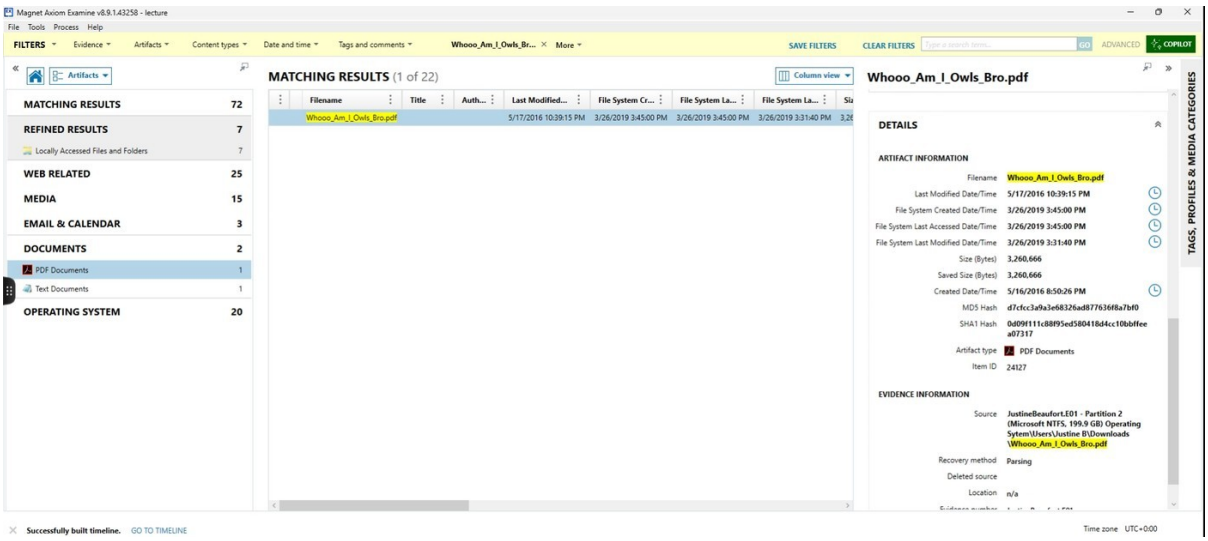


Figure P4-09. *Downloads copy of the PDF (Item ID 24127): file system created 3:45:00 PM, accessed 3:45:00 PM, modified 3:31:40 PM; same MD5 and SHA1 as the emailed attachment.*

7 Limitations

- The examination used AXIOM Examine on a pre-processed case. No evidence was acquired, and findings could not be re-derived from the raw image beyond the hash search.
- Artifacts attribute activity to a user account (Justine B), not to a person at the keyboard. Training data cannot show intent.
- Browser history rows can repeat for one event. Some records were recovered by carving (for example WebCacheV01.dat and the `F:\` root shortcut), which can be less reliable.
- Document properties are written by authoring software, can be altered, and do not state a time zone.
- Last-accessed values on the USB shortcuts (4:00:00 AM) look like date-only values; the file system of the drive was not confirmed.
- Program names for Jump List and Prefetch records are AXIOM's inference from application IDs.
- Not examined: Connections graphs, the Locally Accessed Files and Folders records, the Word file's Carved Archives and Media items, the 13 email items returned by "Great Horned", and the individual Edge/IE history entries for F:/.
- The registry-record discrepancy (6.1) and the 3:45:00 PM time (6.2) are unresolved.

8 Conclusion

The artifacts show two documents downloaded through Chrome on 26 March 2019, each shortly after a file-type search, and a copy of the first created on a removable drive while it downloaded. The same PDF was emailed to Jim Turk about 17 minutes later, and the attachment is identical in content to the file on disk. The removable drive was used for further files until at least 4:49 PM, although its own registry record covers only a short session earlier in the day. These findings are verified by at least two independent artifacts wherever the case allowed. They describe account activity and timing; they do not identify who was using the computer or why, and the two discrepancies in section 6 remain open.

Appendix A Supporting figures

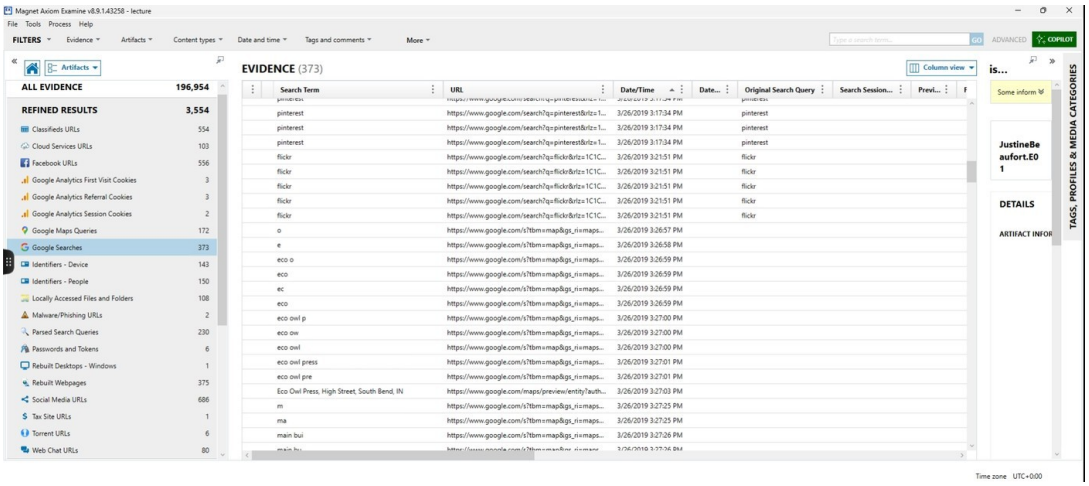


Figure P2-01. *Google Searches, about 3:17 to 3:27 PM (pinterest, flickr and Maps lookups).*

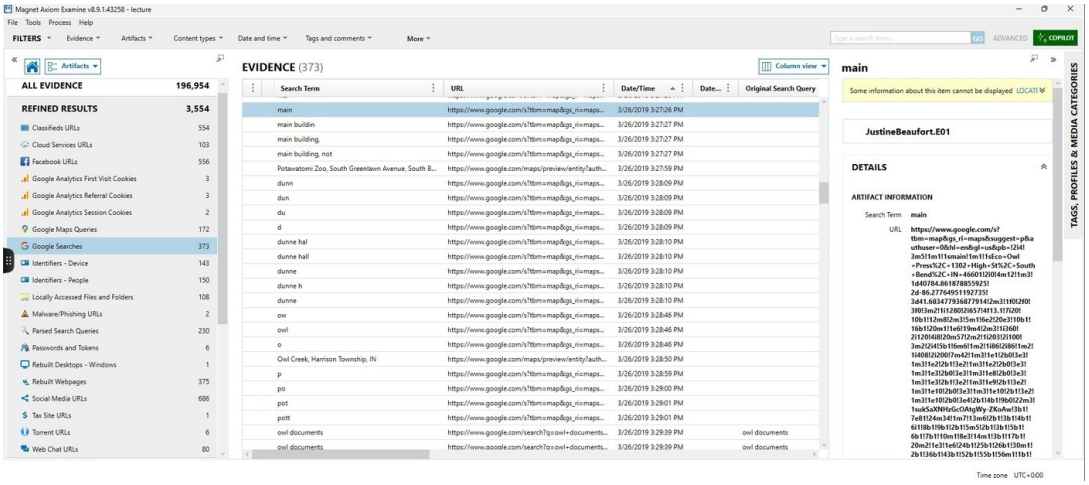


Figure P2-01b. *Google Searches, 3:27:26 to 3:29:39 PM; shows the sequence is continuous between P2-01 and P2-02.*

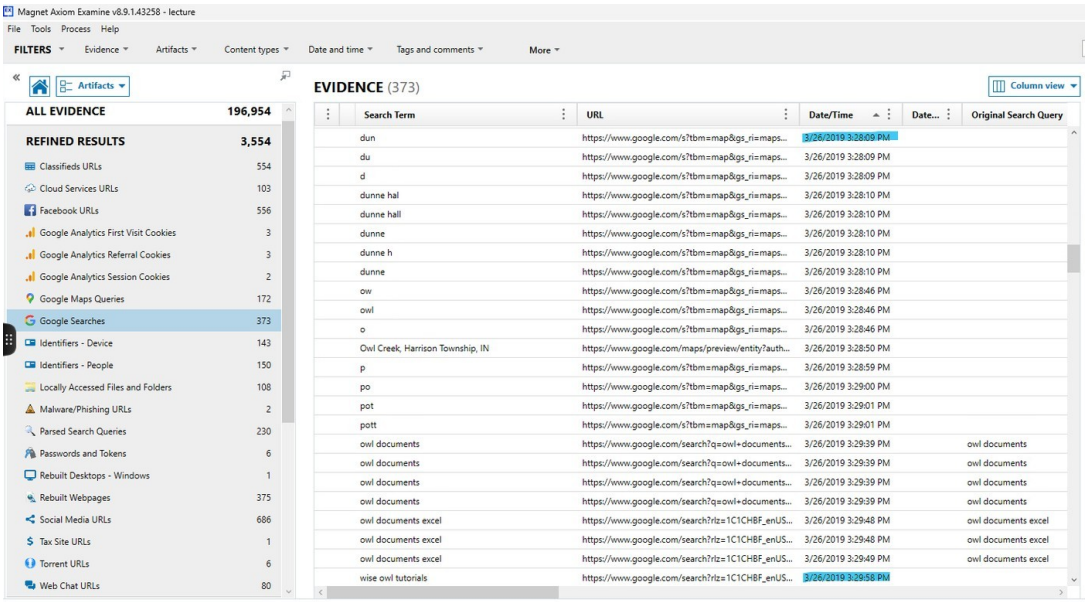


Figure P2-02. *Google Searches, 3:28:09 to 3:29:58 PM (Maps lookups, then owl documents, owl documents excel and wise owl tutorials).*

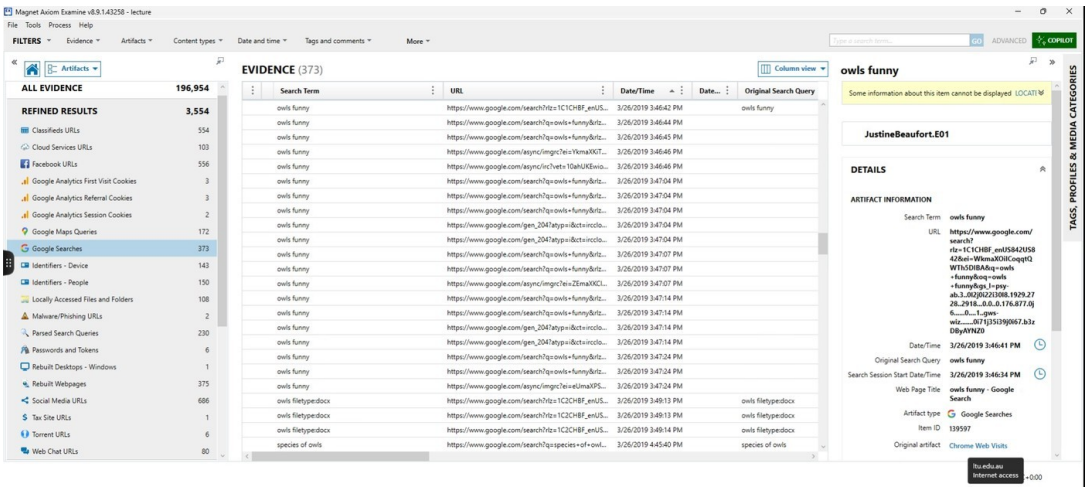


Figure P2-04. *Google Searches, 3:46:42 PM to 4:45:40 PM, including "owls filetype:docx" at 3:49:13 PM.*

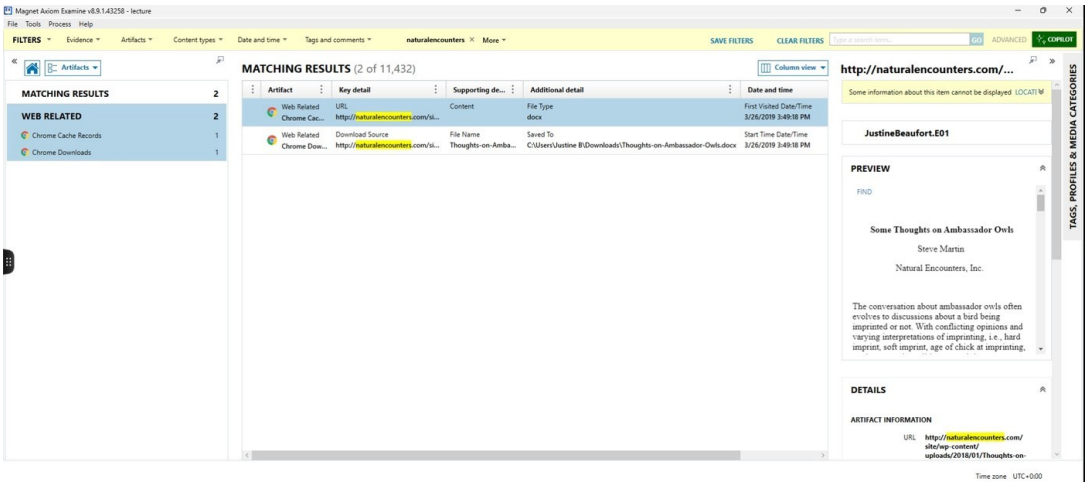


Figure P2-06. *Chrome records for naturalencounters.com: cache entry and download record, both starting 3:49:18 PM; no separate visit record.*

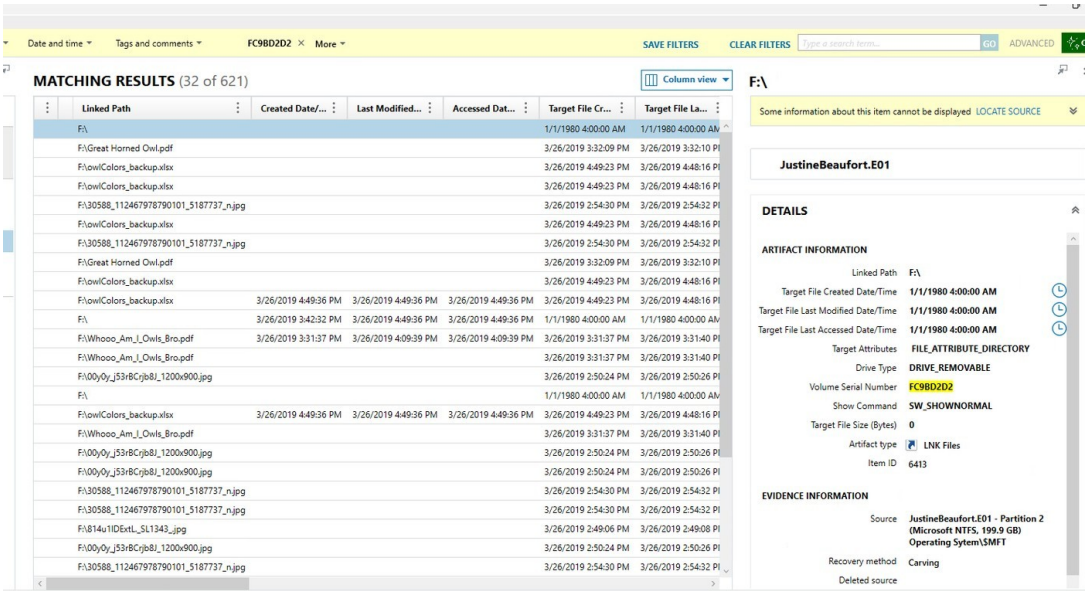


Figure P3-02. *LNK Files matching volume FC9BD2D2 (earliest rows).*

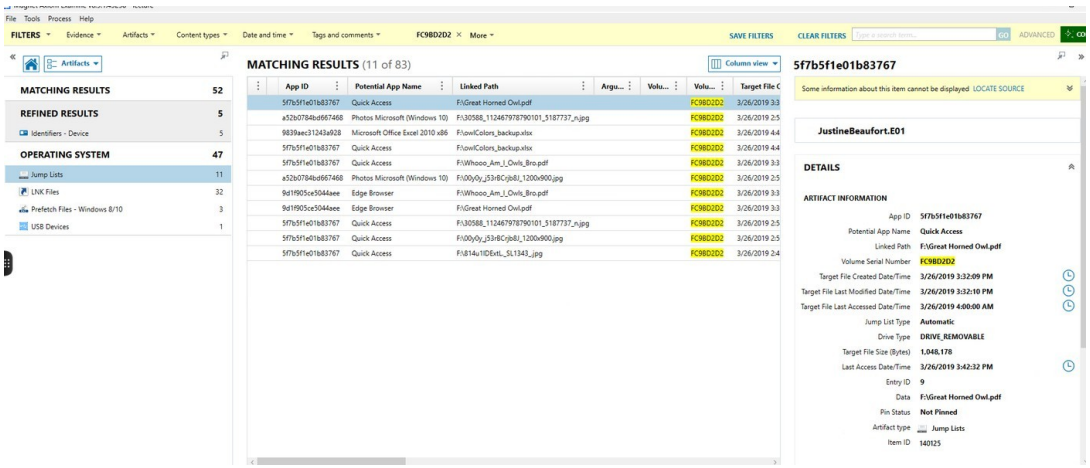


Figure P3-04. *Jump Lists matching volume FC9BD2D2: Quick Access, Edge, Excel 2010 and Photos entries for files on F:\.*

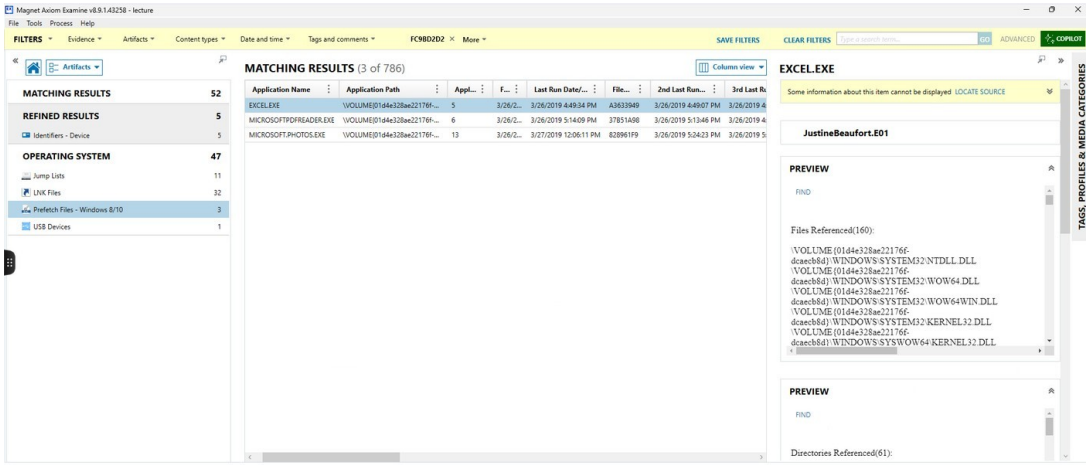


Figure P3-05. *Prefetch Files referencing the volume: EXCEL.EXE (last run 4:49:34 PM), MICROSOFTPDFREADER.EXE (5:14:09 PM), MICROSOFT.PHOTOS.EXE (3/27/2019 12:06:11 PM).*

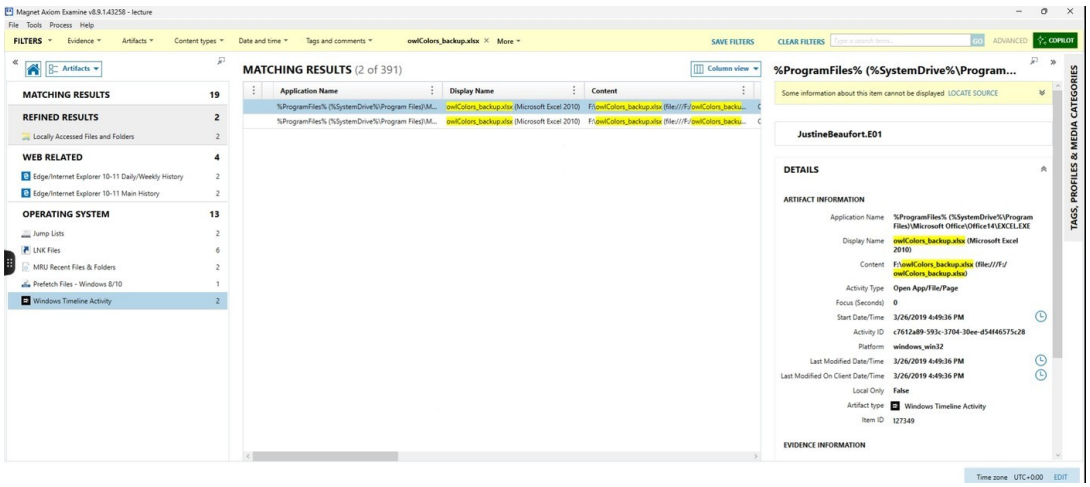


Figure P3-06. *Windows Timeline Activity: Excel 2010 opening F:\owlColors_backup.xlsx at 3/26/2019 4:49:36 PM.*

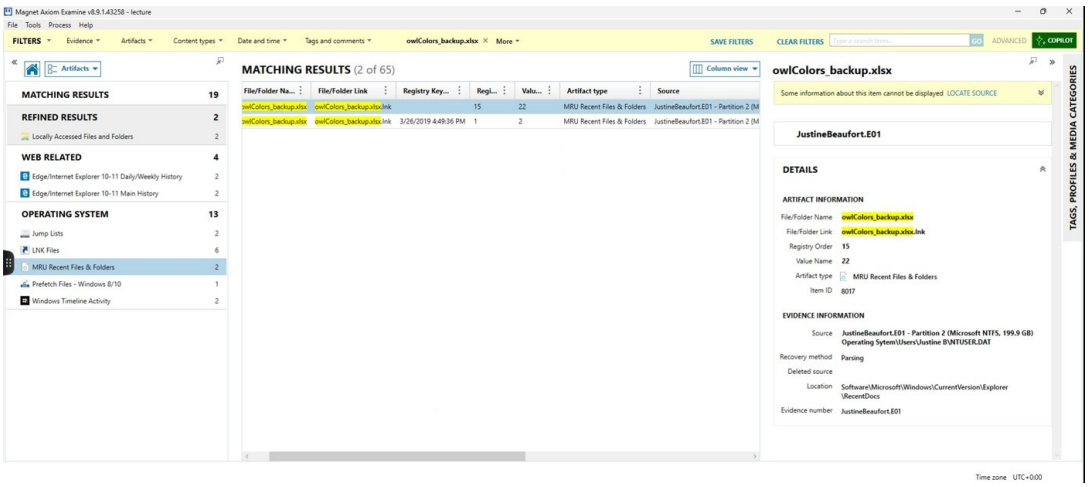


Figure P3-07. *RecentDocs (NTUSER.DAT, Justine B): owlColors_backup.xlsx.lnk with a timestamp of 4:49:36 PM.*

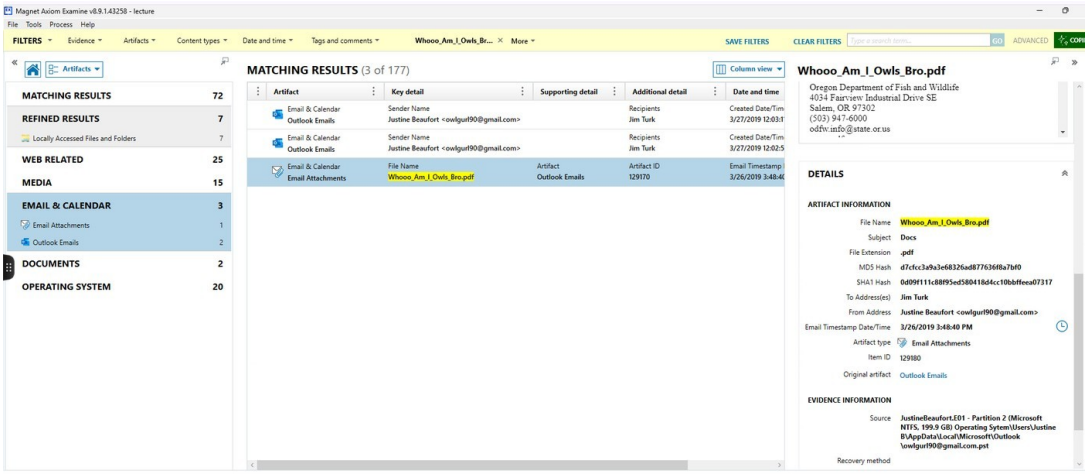


Figure P4-01. *Email & Calendar results for Whooo_Am_I_Owls_Bro.pdf: two Outlook Emails (Sent Mail and All Mail copies) and one Email Attachments record.*

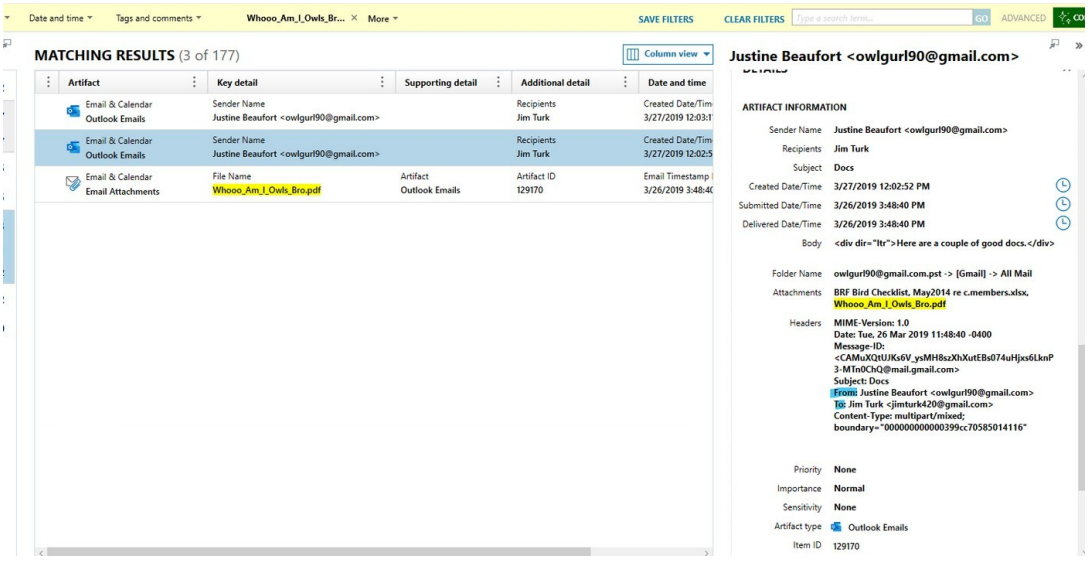


Figure P4-03. *All Mail copy of the same message (same Message-ID, subject and attachments).*

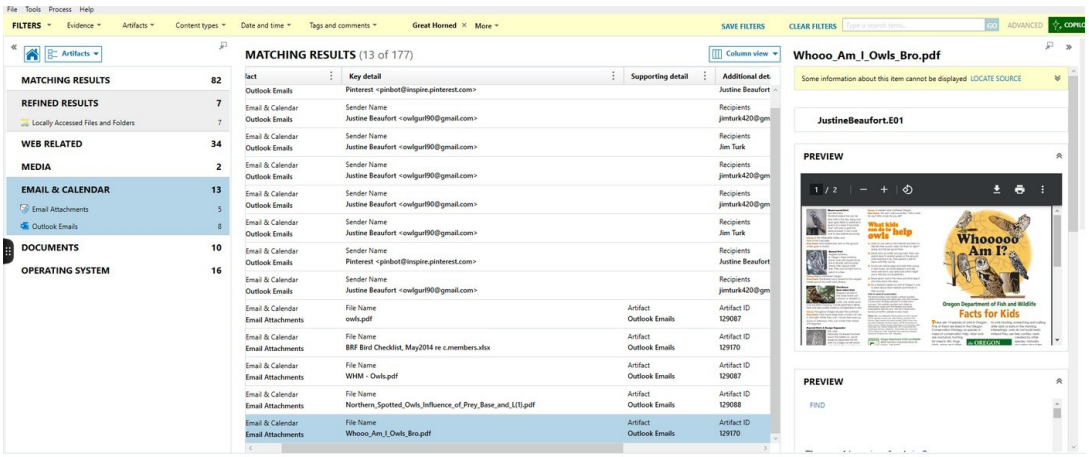


Figure P4-05. *Search for "Great Horned": 13 Email & Calendar items (8 Outlook Emails, 5 attachments); no attachment named Great Horned Owl.pdf.*

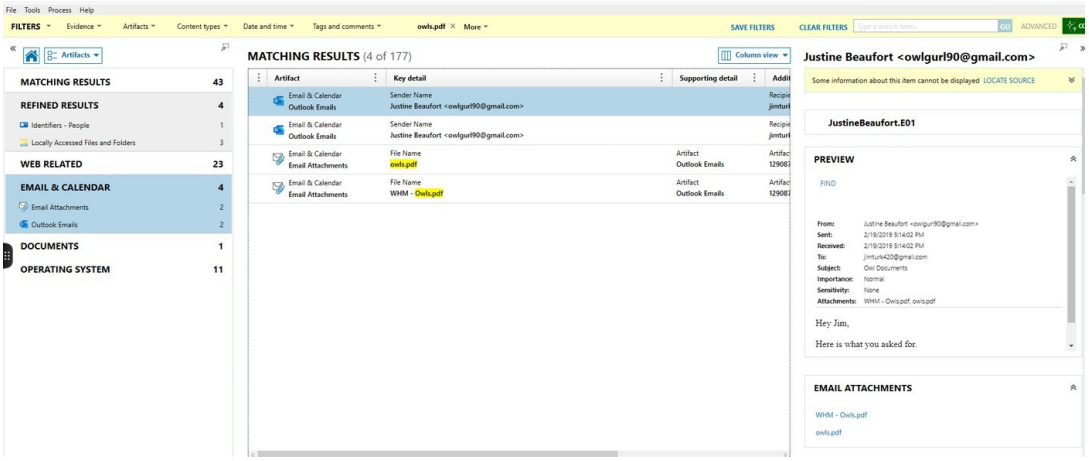


Figure P4-06. *Search for owls.pdf: result categories (no Cloud category) and the preview of the 2/19/2019 5:14:02 PM email "Owl Documents" to jimturk420@gmail.com.*

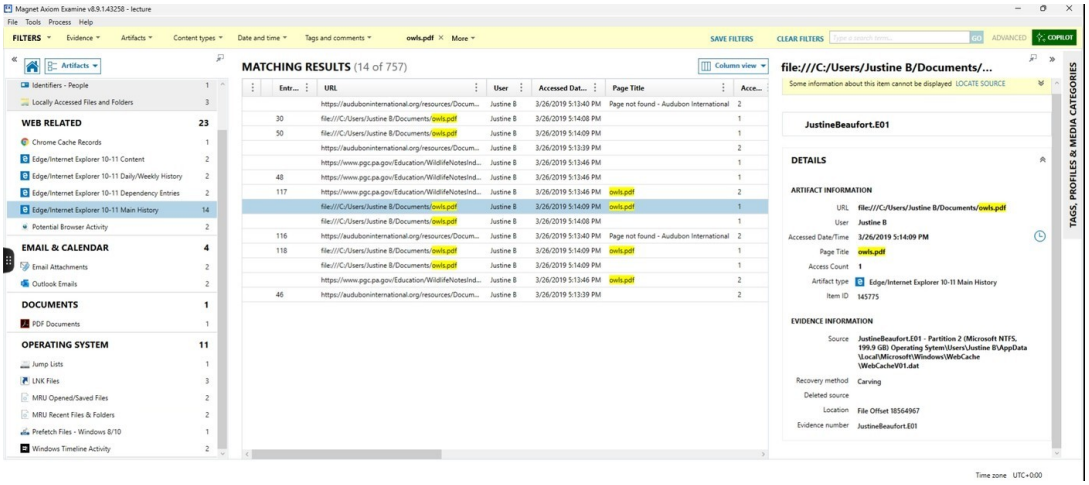


Figure P4-06b. *Search for owls.pdf: Edge/IE history entry for file:///C:/Users/Justine B/Documents/owls.pdf accessed 3/26/2019 5:14:09 PM (carved record).*

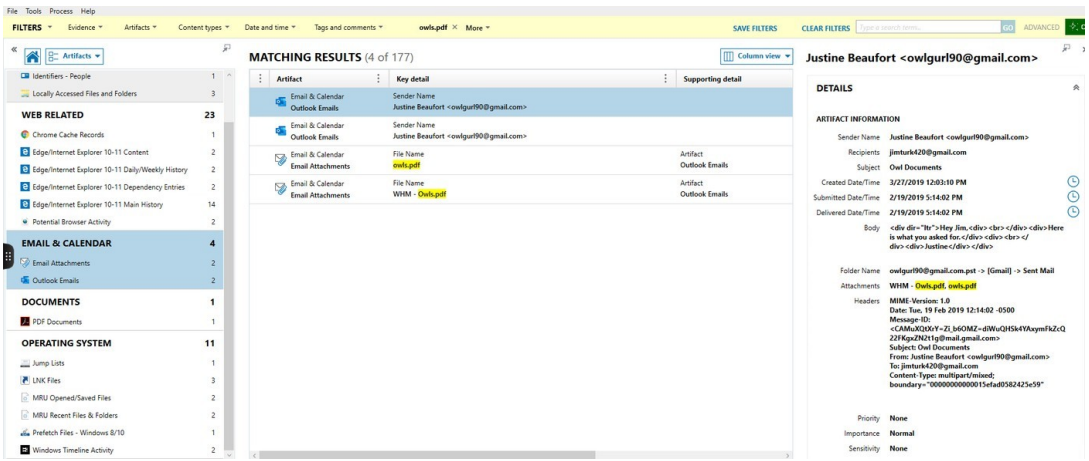


Figure P4-07. *Details of the same 2/19/2019 email (Sent Mail; submitted and delivered 5:14:02 PM) with owls.pdf and WHM - Owls.pdf attached.*

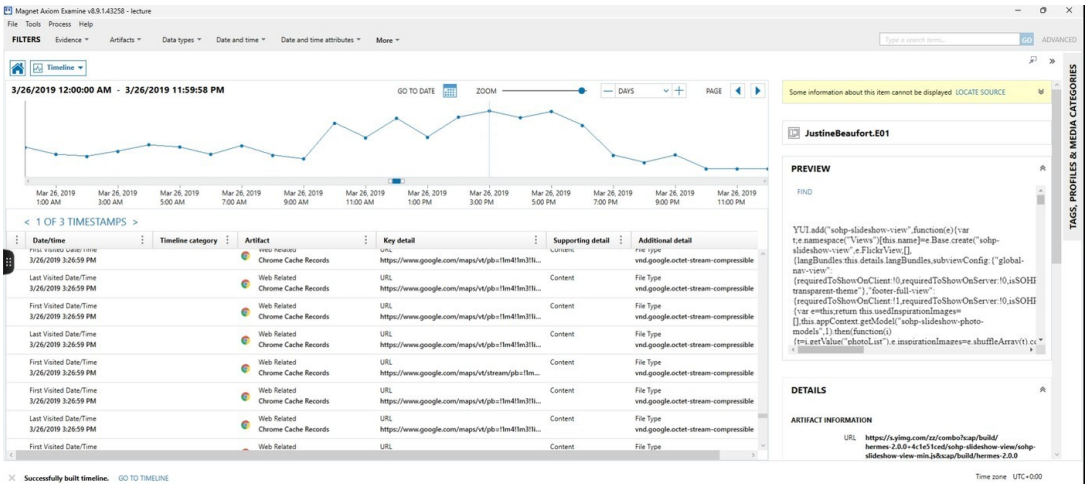


Figure P5-00. *Timeline built in Examine (zoomed to days); the 3:31 PM cluster is shown in Figure P5-01.*